

The following is a guide for reproducing the statistical analyses associated with this data package entitled “Data and scripts associated with a manuscript investigating levoglucosan dynamics in fire-affected watersheds in coastal California, USA ”

Doi:

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Make sure to follow this guide step-by-step in order to reproduce figures and statistics associated with this project.

Download Necessary Files:

- 1) Download all scripts and data from the ESS-DIVE repository (Barton et al. 2025; **doi:** add the data package one)

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a. Scripts:

- i. “CoastalFires_DataFrameWrangling.Rmd”
- ii. “CoastalFires_DataAnalysis.Rmd”

b. Data:

- i. “CoastalFires_LevoglucosanData.csv”

- 2) Download associated data from CUAHSI Hydroshare. New data (levoglucosan data) is published with this file and the scripts associated with the manuscript. However, data pertaining to the watersheds, discharge, and black carbon concentrations have been previously published in association with a sister paper titled: “Hydrology, rather than wildfire burn extent, determines post-fire organic and black carbon export from mountain rivers in central coastal California”.

Citation: Barton, R., C. Richardson, E. Pae, M. Montalvo, M. Redmond, M. Zimmer, S. Wagner (2025). Hydrology, rather than wildfire burn extent, determines post-fire organic and black carbon export from mountain rivers in central coastal California [Dataset], HydroShare, <https://doi.org/10.4211/hs.421c0226bb38460c8393d67fe0c4f802>

a. Data Files:

- i. “CoastalFires_BiogeochemData.csv”
- ii. “CoastalFires_DischargeData.csv”
- iii. “CoastalFires_WatershedData.csv”

Folder Setup Instructions:

- 1) The manuscript data package will contain the following folders:
 - a. Input
 - b. Output
- 2) From this data package, the “Input” folder should already contain the levoglucosan data file named: “Coastal_Fires_LevoglucosanData.csv”. Within this folder, insert the CUAHSI Hydroshare files named:
 - a. “CoastalFires_BiogeochemData.csv”
 - b. “CoastalFires_DischargeData.csv”
 - c. “CoastalFires_WatershedData.csv”

Initial Data Processing Instructions:

- 1) In order to combine the datasets and process data prior to data analysis, run the following script:
 - a. “CoastalFires_DataFrameWrangling.Rmd”

*Notes: The script is written in R-Markdown chunks that must be completed in order. This data processing is required before running the DataAnalysis script. We have already run these scripts. The output of this script is a file titled “CoastalFires_CombinedData.csv” which is utilized for further data analysis and figure production and can be found in the folder titled “Output.”

Data Analysis:

- 1) In order to conduct all statistical analyses associated with the manuscript, run the following script:
 - a. Run “CoastalFires_DataAnalysis.Rmd”
- 2) This will result in a variety of statistical outputs as well as figures showcasing the results of the NMDS. Both a labeled and unlabeled NMDS figure will be found in the “Output” folder titled:
 - a. “NMDS_unlabeled.png”
 - b. “NMDS_labeled.png”

*Note: The script is written in R-Markdown chunks that must be completed in order.